

A Two-Stage Optical Design Methodology for an Eyebow-Expanded 3D AR-HUD

Junnan Jin, Jing Zhang, Yuan Yuan, Jiangbo Yao, Zhifu Li, Bin Zhao

TCL China Star Optoelectronics Display Technology Co. Ltd, Shenzhen, Guangdong, China

Abstract

This paper presents a systematic and efficient two-stage optical design methodology for developing 3D Augmented Reality Head-Up Displays (AR-HUDs) with an expanded eyebow and suppressed crosstalk. Conventional design approaches struggle to balance these competing performance metrics due to their computational inefficiency and high sensitivity to initial parameter selection. To overcome these limitations, we introduce a dual-phase design strategy: first, a computationally efficient slit-mask model enables rapid parameter space exploration to identify optimal viewing zone configurations for eyebow maximization; subsequently, these results are translated into precise initial parameters for high-fidelity lenticular lens simulation and optimization in the 3D AR-HUD system. A prototype developed using this methodology achieves an expanded horizontal eyebow of ± 65 mm while maintaining crosstalk below 5%, demonstrating excellent agreement with simulation results. These findings validate the proposed approach as both feasible and efficient for developing high-performance 3D AR-HUDs, establishing a systematic, replicable technical framework for future implementations.

Author Keywords

3D Display, AR-HUD, Lenticular Lens, Eyebow, Optical Design, Simulation

1. Introduction

AR-HUDs enhance driving safety and interaction intuitiveness by projecting critical information as virtual images onto the windshield, seamlessly integrating with the real-road environment [1]. The ongoing transition from 2D to 3D display technology aims to further improve situational awareness through depth cues, enabling more precise navigation guidance and collision warnings [2].

Among autostereoscopic 3D solutions, the lenticular lens approach has emerged as a promising candidate for automotive applications, owing to its compact form factor, high optical efficiency, mature fabrication processes, and favorable cost structure [3]. This technology directs light from interleaved left- and right-eye pixel columns into distinct viewing zones, allowing drivers to perceive stereoscopic imagery without auxiliary eyewear.

Nevertheless, two key performance metrics of lenticular 3D systems—eyebow (the spatial region where correct stereoscopic images are visible) and crosstalk (undesired interference between left- and right-eye channels)—exhibit high sensitivity to lens parameters such as pitch and focal length [4]. Conventional design methodologies typically perform direct optimization on full lenticular models, a computationally intensive process prone to local optima convergence and strong dependence on initial parameter guesses. These limitations often result in protracted development cycles and suboptimal efficiency.

To address these challenges, this paper introduces a systematic two-stage optical design methodology. The core concept employs

a structurally simplified slit-mask model as an optical proxy for the lenticular lens. During the first stage, rapid parametric scanning identifies globally optimal performance regions for eyebow and crosstalk. These results are then mapped into high-precision initial values for the second stage—high-fidelity lenticular lens simulation and optimization. This structured approach enables seamless transition from system-level exploration to component-level design, effectively decoupling different hierarchical tasks to significantly enhance overall design efficiency and optimization reliability. factors.

2. Two-Stage Simulation Design Methodology

2.1 Stage 1: Rapid Parametric Exploration Using Slit-Mask Model

This study employs a slit-mask as a geometrically equivalent model for the initial optical design of lenticular lenses. Although lenticular lenses operate on refractive principles to converge light while slit-masks utilize physical occlusion for ray selection, both systems share highly similar geometric relationships in viewing zone formation [5]. This fundamental similarity establishes the slit-mask as an effective proxy for preliminary design.

The optical system is constructed as follows: light emitted from the micro-display is modulated by the slit-mask, then sequentially passes through the secondary mirror, primary mirror, and windshield, ultimately forming viewing zones at the eyebow plane (see Fig. 1 for a schematic diagram). To achieve optimal performance, we first established a comprehensive slit-mask optical model and enhanced its accuracy by optimizing pixel emission angles and luminous region characteristics. Subsequently, we performed parametric verification and optimization of the key design parameters: slit pitch and its distance from the virtual display plane.

The primary objective of this stage is to systematically conduct parametric sweeps, evaluate the theoretical horizontal eyebow size under different parameter combinations, and identify the configuration that maximizes the eyebow range. The optimized parameters from this stage serve as refined initial inputs for the detailed lenticular lens design in the second stage. be achieved.

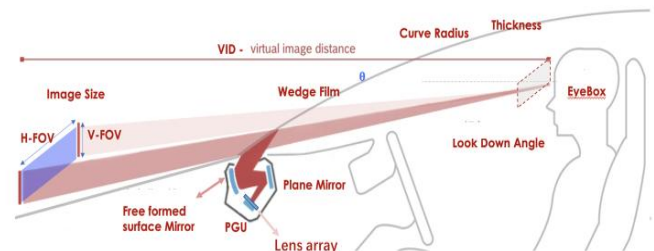


Figure 1. Schematic diagram of the optical system

2.2 Stage 2: Precision Design and Fabrication of Lenticular Lens

Building upon the optimized parameters derived from the first-stage slit-mask model, this phase translates these findings into the physical design space of the lenticular lens. Using paraxial optical equations, the optimal slit pitch (p) and mask-to-display distance (d) are converted into initial structural parameters for the lenticular lens, comprising lens pitch (p), radius of curvature (r), and sag height (h).

A high-fidelity lenticular lens optical model was subsequently developed and subjected to refined simulations with multi-parameter co-optimization. To enhance manufacturing feasibility, comprehensive tolerance analysis was implemented, ensuring the robustness of the optical solution under practical production conditions. This phase culminates in a complete set of well-defined, manufacturable optical parameters for the lenticular lens, providing reliable foundation for subsequent prototype development.

3. Results and Discussion

3.1 Analysis of Grating Simulation and Experimental Results

Parameters for Grating Simulation as shown in table 1 and 2.

Table 1 Display Module Parameters

Screen	Proposal
screen size (inch)	5.5
resolution	1440*RGB*480
pixel size (μm)	85.35
RGB arrangement	Strip
CF glass thickness (mm)	0.4
Pol. Type	Nitto
Pol. thickness	POL: 117 μm (CF) POL: 111 μm (TFT)
surface treatment	Hard coating

Table 2 HUD Projection Unit Parameters

HUD	Proposal
Interpupillary distance	65mm
Eyebox	140*55 mm
FOV	15°*5°
VID	6.3m
Horizontal magnification factor	12.85
Vertical magnification factor	13.33
Projected eyebox dimension	Computed based on scaling relationships
eyebox	72.9*31.4 mm
Image source to eyebox distance (H)	243.6 mm

The emission model implemented in LightTools software is presented in Fig. 2(c), showing a 780 μm separation between the RB pixel emission layer and the grating structure. In Fig. 2(b), the notation $n=1$ indicates that the space between the grating and the emission layer is modeled as a medium with a refractive index of 1.0 (air/vacuum).

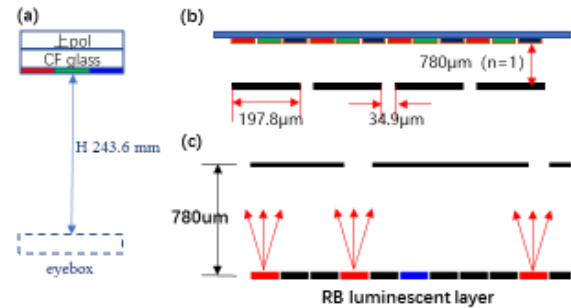


Figure 2. (a) Schematic of the display-to-virtual-eyebox distance; (b) Grating parameter calculation diagram; (c) Pixel emission modeling illustration.

Figure 3 illustrates the schematic pixel arrangement, where approximately eight subpixels are positioned beneath each grating unit. To achieve the stereoscopic effect, all green (G) subpixels are maintained in an "off" state. The red (R) subpixels located to the left of the slit grating center are activated ("on" state), while the blue (B) subpixels to the right of the center are turned on, thereby creating red-blue color separation for 3D perception.

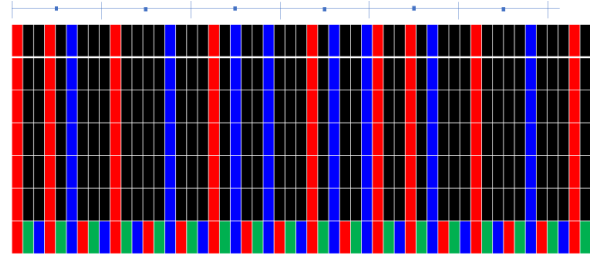


Figure 3. The schematic pixel arrangement

Simulation results confirm that the designed slit grating exhibits favorable performance in both eyebox range and red-blue separation. Figure 4 illustrates the color separation patterns and corresponding energy distributions under different emission angles of the RB pixels. Comparative analysis of Figs. 4(a), (c), and (e) reveals that the system achieves its maximum eyebox of approximately ± 65 mm when the RB pixels' emission angle is set to $\pm 15^\circ$.

Figure 5 further demonstrates the color separation effects and energy distributions with the light source positioned at different locations on the display. As shown in Fig. 5(a), when the source is centered above the display, the red and blue regions are correctly positioned on the left and right sides, respectively, aligning with theoretical expectations. When the source shifts leftward, both color regions move correspondingly to the left; conversely, a rightward source displacement causes both regions to shift right. These results indicate that the designed slit grating structure achieves excellent color separation, with response

behavior consistent with practical requirements, thereby validating the rationality and reliability of the design.

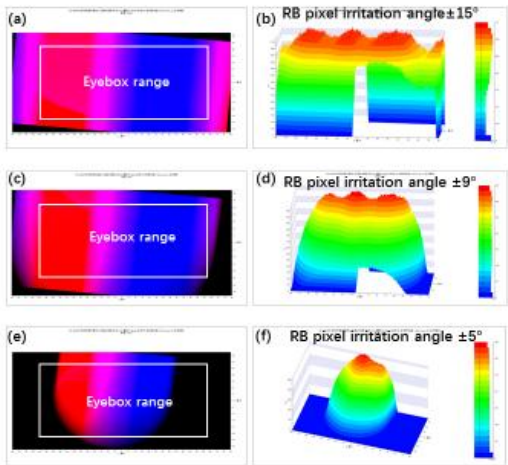


Figure 4. Red-blue separation patterns and energy distribution under different emission angles of RB pixels.

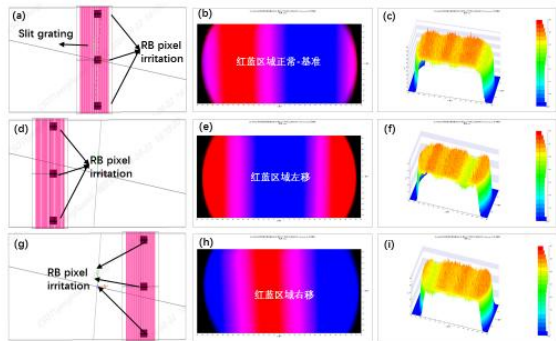


Figure 5. Red-blue separation performance and energy profile with varying source positions across the display.

Figure 6 presents the normalized red-blue crosstalk results for the designed slit grating parameters. The measured crosstalk remains at a low level of approximately 4.8%, demonstrating excellent red-blue separation performance of the proposed grating structure.

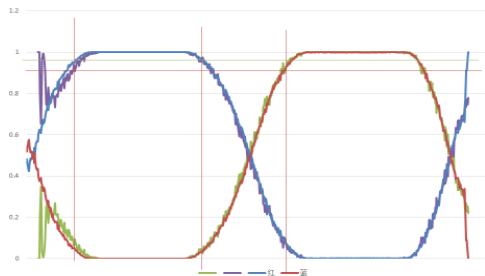


Figure 6. The normalized red-blue crosstalk results

Figure 7 presents the experimental validation results of the slit grating design. These results further confirm that the grating with the designed parameters effectively expands the eyebox while maintaining satisfactory red-blue separation performance.

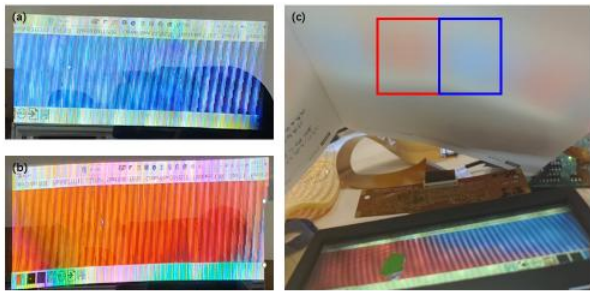


Figure 7. The experimental validation results of the slit grating design

3.2 Lenticular Lens Array Simulation and Results Analysis

Based on the simulation results from the slit grating model, the display and optical engine parameters for the 3D AR-HUD with lenticular lenses were optimized. The specific design parameters are summarized in Tables 3 and 4.

Table 3 Display Module Parameters

Screen	Proposal
screen size (inch)	5.5
Transmission	Avg.6.2%
Aspect ratio	16: 9
resolution	3840*RGB*2160
pixel size (mm*mm)	31.8*31.8
A/A Size (mm*mm)	122.1 (H) *68.6 (V)
Outline (mm)	125.1*74.9
RGB arrangement	Strip
CF glass thickness (mm)	0.4
Pol. Type	Nitto
Pol. thickness	POL: 115μm (CF) POL: 128 um (TFT)

Table 4 HUD Projection Unit Parameters

HUD	Proposal
Interpupillary distance	65mm
Eyebox	130*55 mm
FOV	15°*5°
VID	6.3m
Horizontal magnification factor	12.85
Vertical magnification factor	13.33
Projected eyebox dimension	Computed based on scaling relationships
eyebox	67.7*31.4 mm
Image source to eyebox distance (H)	243.6 mm

Figure 8 illustrates the assembly relationship between the lenticular lens and the display panel. To mitigate moiré patterns, the lenticular lens is tilted 14° clockwise relative to the display screen during lamination.

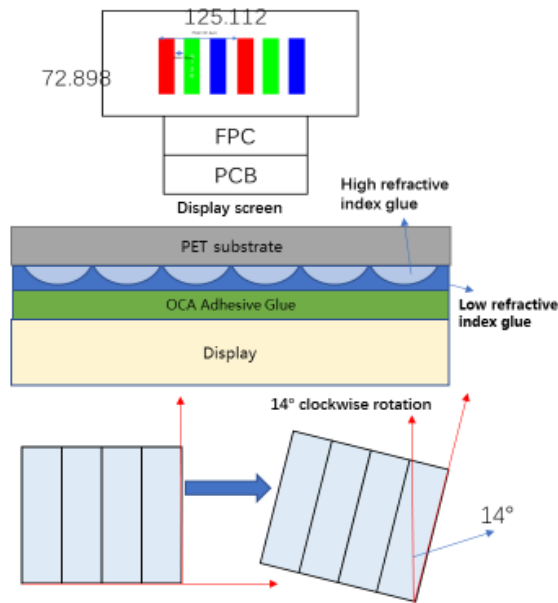


Figure 8. The assembly relationship between the lenticular lens and the display panel.

Figure 9 illustrates the designed pixel arrangement architecture. In this configuration, the lenticular lens array is oriented with a 14° clockwise tilt, with each lens period corresponding to a "red-left, blue-right" subpixel organization. To enhance image quality, subpixel units exhibiting excessive crosstalk have been selectively disabled. The simulation domain encompasses approximately 13 lenticular periods horizontally and 10 pixel periods vertically, providing comprehensive evaluation of the structure's optical performance at practical scales.

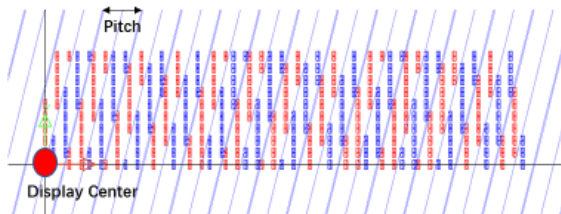


Figure 9. The designed pixel arrangement architecture

Figure 10 presents the simulation results of the lenticular lens grating with the designed parameters, along with experimental results obtained after fabrication and screen lamination. Specifically, panels (a) and (b) display the simulated red-blue separation patterns and corresponding energy distributions, respectively. Comparative analysis between simulated predictions and measured data demonstrates that the fabricated lenticular lens grating achieves excellent color separation capability, characterized by low crosstalk and an expanded eyebox range, thereby validating the effectiveness of the design parameters.

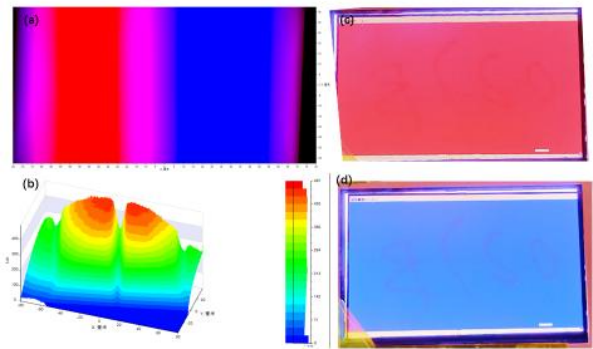


Figure 10. Lenticular Lens: Simulation vs. Experimental Results

The primary advantage of this approach lies in its substantially improved design efficiency. By employing the slit grating as a rapid proxy, extensive design space exploration was achieved with minimal computational resources, effectively avoiding the local optima traps common in direct lenticular lens optimization. This methodology establishes a more deterministic and reliable pathway from concept to high-performance implementation.

A limitation of this study is its primary focus on horizontal parallax. Future work will extend this methodology to hybrid lens designs to achieve simultaneous expansion of both horizontal and vertical eyebox dimensions.

4. Conclusion

This paper has successfully proposed and validated a novel dual-stage optical design methodology for developing lenticular lens-based 3D AR-HUD systems. The implemented approach, which utilizes a slit-mask proxy for rapid system-level optimization followed by high-fidelity lenticular refinement, has demonstrated both computational efficiency and predictive accuracy. A functional prototype developed using this methodology achieves an expanded eyebox of ± 65 mm with maintained low crosstalk ($< 4.8\%$), confirming its practical utility. The established framework provides a robust foundation for significantly accelerating the design cycle and performance optimization of 3D AR-HUDs, thereby facilitating their broader adoption in next-generation vehicles.

5. References

- [1] C. Böke, et al., "Augmented Reality in the Cockpit: A Review of HUD and HMD Technology," *SID Symposium Digest*, vol. 52, pp. 312-315, 2021.
- [2] J. Geng, "Three-dimensional display technologies," *Adv. Opt. Photonics*, vol. 5, no. 4, pp. 456-535, 2013.
- [3] N. A. Dodgson, "Autostereoscopic 3D displays," *Computer*, vol. 38, no. 8, pp. 31-36, 2005.
- [4] H. Huang, et al., "Analysis of the viewing zone of lenticular sheet-based autostereoscopic displays," *J. Soc. Inf. Display*, vol. 24, no. 4, pp. 245-251, 2016.
- [5] S. -W. Min, et al., "Analysis of an orthographic viewing zone in lenticular 3D displays," *Opt. Express*, vol. 15, no. 20, pp. 13333-13341, 2007.